

Angles & Angle Relationships

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Standard: CCSS.MATH.CONTENT.7.G.B.5 — Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle.

Problem 1 — Vocabulary Match

Match each term (Point, Line Segment, Ray, Vertex, Angle) with the correct definition (A–E).

Vocabulary

Geometry Basics

- 1a. **Point** → C (a location with no length, width, or height)
- 1b. **Line Segment** → B (part of a line with two endpoints)
- 1c. **Ray** → E (starts at one point, extends forever in one direction)
- 1d. **Vertex** → D (the point where two lines or segments meet)
- 1e. **Angle** → A (a figure formed by two rays sharing a common endpoint)

1a→C, 1b→B, 1c→E, 1d→D, 1e→A

Problem 2 — Name the Angle

An angle has vertex B with rays $\overrightarrow{BA}$ and $\overrightarrow{BC}$. Write three names for the angle.

Angle Notation

Vertex Rule

Rule: when naming an angle with three letters, the *vertex must be in the middle*.

- (i) $\angle ABC$ — vertex B in the middle, A on one ray, C on the other.
- (ii) $\angle CBA$ — same angle, just listed the rays in the opposite order.
- (iii) $\angle B$ — short name using just the vertex letter (allowed only when there is no other angle at the same vertex to confuse it with).

$\angle ABC, \angle CBA, \angle B$

Problem 3 — Classify 35°

$\angle 1$ measures 35° . Classify as acute / right / obtuse / straight.

Angle Classification

Acute: $0^\circ < \text{angle} < 90^\circ$

35° is between 0° and 90° , so it is **acute**.

Acute

Problem 4 — Classify 90°

$\angle 2$ measures 90° . Classify.

Angle Classification

Right: exactly 90° .

$90^\circ = 90^\circ$, so it is a **right angle**. (Look for the small square mark in figures.)

Right

Problem 5 — Classify 142°

$\angle 3$ measures 142° . Classify.

Angle Classification

Obtuse: $90^\circ < \text{angle} < 180^\circ$

142° is between 90° and 180° , so it is **obtuse**.

Obtuse

Problem 6 — Classify 180°

$\angle 4$ measures 180° . Classify.

Angle Classification

Straight: exactly 180° — the two rays form a straight line.

$180^\circ = 180^\circ$, so it is a **straight angle**.

Straight

Problem 7 — Complement of 27°

Two angles are complementary. One is 27° . Find the other.

Complementary Angles

Sum = 90°

Definition: Two complementary angles add up to 90° .

Set up: $27^\circ + x = 90^\circ$

Subtract 27° from both sides: $x = 90^\circ - 27^\circ = 63^\circ$.

63°

Problem 8 — Supplement of 118°

Two angles are supplementary. One is 118° . Find the other.

Supplementary Angles

Sum = 180°

Definition: Two supplementary angles add up to 180° .

Set up: $118^\circ + x = 180^\circ$

Subtract 118° : $x = 180^\circ - 118^\circ = 62^\circ$.

62°

Problem 9 — Vertical Angles

$\angle 1$ and $\angle 2$ are vertical angles, $m\angle 1 = 64^\circ$. Find $m\angle 2$ and explain.

Vertical Angles

Congruent

Key fact: Vertical angles are *always congruent* (equal in measure).

So $m\angle 2 = m\angle 1 = 64^\circ$.

Why: When two lines cross, the angles directly across from each other (vertical angles) are formed by the same pair of opposite rays, so they must have the same measure.

$m\angle 2 = 64^\circ$ (vertical angles are congruent)

Problem 10 — Adjacent + Straight Angle

$\angle ABC$ and $\angle CBD$ are adjacent and form a straight angle. $m\angle ABC = 75^\circ$. Find $m\angle CBD$.

Adjacent Angles

Linear Pair

Sum = 180°

Setup: Two adjacent angles whose non-shared sides form a straight line are a *linear pair*, and their measures add to 180° .

$$m\angle ABC + m\angle CBD = 180^\circ$$

$$75^\circ + m\angle CBD = 180^\circ$$

$$m\angle CBD = 180^\circ - 75^\circ = 105^\circ.$$

$m\angle CBD = 105^\circ$

Reminder for Natalie: When you see a 3-letter angle name like $\angle ABC$, the **middle letter is always the vertex**. For complement / supplement problems, just remember the magic numbers — **complement adds to 90°** , **supplement adds to 180°** . Vertical angles (across the X) are equal; adjacent angles on a straight line add to 180° .